Blackout — play-through review

Theme `blackout` · senior-high electrical engineering (grade 12 manifest) · 15 days, ~45 stops · reviewed 2026-08-21 by reading the full book (`books/blackout.yml`), solving every question, and reading opening, stakes, warm-ups, interiors and ending in play order.

Verdict

Structurally the strongest campaign in the set. Its spine is a single wrong number: on day 4 the room holds a corridor inside an emergency rating derived from a sensor nobody has checked; on day 9 that corridor trips at the worst hour of the worst day; on day 11 Farrow traces the sensor and the rating was never real; day 12 re-reads the whole fortnight from it; day 15 makes "cross-check any instrument a decision rests on" the practice that outlives the emergency. Days 9–12 are the best four consecutive missions in the repo. The two CASEBOOK stops on day 11 — which of the week's conclusions still stand, and why each one falls or holds — are the single most sophisticated thing any of these games asks a player to do.

Against that: **three CHOICE stops are keyed to the wrong option**, two of them in this book and one in its junior edition, and all three are live in the shipped generated content. On each, the stop's own `answerText`, `why` and rebuttals state the correct answer while `answer:` names a distractor — so a player who reasons correctly is marked wrong and then shown a verdict that agrees with them. One of these is the transformer stop that `equationSupply.mjs` was written for.

Answerable: 41/45. Three stops mark the right answer wrong (BL-01, BL-02, BL-03); the corridor's numbers do not reconcile across five stops (BL-04). **Sense:** Excellent between missions. Inside individual cards, this book has the most numeric drift of any reviewed. **Level:** Right, and the ladder is well built — I^2R, then three-phase $\sqrt{3}$, then impedance in quadrature, then a fault current, then annealing. **Fun:** High, and unusually well paced: day 10 is deliberately a shift where nothing happens, and the book says so.

Opening blurb

> "*The Calder network carries electricity to four million people, and all of it runs on one number… A fault takes seconds; getting a system back takes days.*"

Both leads named with their positions and their disagreement stated. One of the two best openings in the set. Keep.

The questions, solved

Verified: $187/220 = 0.85$; $3 \times 1150^2 \times 4.2 = 16.66$ MW; $3 \times 1280^2 \times 4.2 = 20.6$ MW; $2 \times 25{,}000 \times 0.30 \div 50 = 300$ MW; $\sqrt{(3.1^2 + 12.4^2)} = 12.78\ \Omega$; $900 \times 3 \div 9 = 300$ A; $1.732 \times 11{,}000 \times 310 \times 0.95 = 5.61$ MW; $\sqrt{3} \times 310 \times (1.8{\times}0.95 + 0.9{\times}0.31) = 1{,}069$ V; $6{,}350 \div 0.42 = 15.1$ kA; $6{,}400 + 120{\times}4 = 6{,}880$ MW; $300 \times 2.5 = 750$ MWh. The DEGENERACY (island: $|\Delta P|/E\_k = 0.004\ s^{-1}$, inertia measured at 25,000 MW·s ⇒ 100 MW) is correct and is the best use of that format in the repo — a slope genuinely cannot separate the two unknowns. The CHAIN's governing link is the 8 MW startup feeder against a 500 MW distractor. The SWEEP's curve steepens correctly (heat ∝ I^2).

Findings

> ***Closed on 2026-08-22.*** *The rows marked CLOSED below were fixed and verified after the first pass; see* `_rejudged.pdf` *for the re-read.*

ID	Sev	Where	Issue	Proposed fix
BL-01	**FIX**	Day 1 "The reason the wires are not at wall voltage" (CHOICE)	The key is wrong. Options include "Current falls by 20× and resistive line loss falls by **400×**" (correct) and "…falls by **20×**", and `answer:` names the 20× one. The stop's `answerText` says "Resistive loss then falls by 20^2 = 400×", the `why` says "Resistive loss then falls with current squared", and the first rebuttal — written against a wrong option — says "Loss then scales with the square of that current, **not in direct proportion**", which refutes the keyed answer. Verified live: `themes/blackout/content/curriculum.js` carries `correctChoice: "…falls by 20×"`. This is the stop CLAUDE.md records as the origin of `equationSupply.mjs`.	One line: `answer: Current falls by 20× and resistive line loss falls by 400×.` Then re-import and confirm `correctChoice` in the generated content.
BL-02	**FIX**	Day 6 "Two instruments, two quantities" (CHOICE)	Same defect. Options are `325 V ≈ 230 V × √2` (correct, first) and `325 V ≈ 230 V × 2`; `answer:` names the ×2 option. `answerText` says "325 V ≈ 230 V × √2", the `why` derives 325/1.414 = 230, and the first rebuttal says "Doubling would make peak twice RMS, but a sine wave differs by root two." `books/blackout-fable.yml` **— the same-grade rewrite of this campaign — keys the identical stop correctly**, which is what proves this is a regression rather than a choice.	`answer: 325 V ≈ 230 V × √2.`
BL-03	**FIX**	`blackout_ms` day "Two instruments, two numbers" (CHOICE)	The junior edition has the same bug in the same lesson: options include "The peak is about 1.4 times the meter's number" (correct) and "about twice"; `answer:` names "about twice", and the first rebuttal reads "Twice would make the peak 460, and the scope is showing 325." Separately, that stop's `why` says "the peak sits about **one and a half** times higher" — the factor is 1.41, and 1.5 is a number a grade-6 reader will carry forward.	`answer: The peak is about 1.4 times the meter's number.` and change "one and a half times higher" to "about one and a half times" → better, "about forty per cent higher".
BL-04	CLOSED	The corridor, across five stops and one interior panel	The corridor's numbers do not form one picture. Continuous rating is **1,060 A** on the TRANS interior panel and **~1,150 A** in the day-11 SWEEP (the current at which the conductor reaches its 75 °C limit). Loading is "108% of	Fix one canonical set and propagate. Suggested, because it preserves the SWEEP reveal and day

			continuous" in the day-4 TRIGGER (= 1,145 A at 1,060, or 1,242 A at 1,150) while day 2 has the corridor carrying **1,280 A** and day 1 has **1,150 A**. Emergency rating is "1,240 A for 30 min" on the panel and "120% of continuous" in the TRIGGER (= 1,272 A at 1,060). A player tracking the corridor — which days 4, 9 and 11 all require — cannot reconcile these.	4's 108%: continuous **1,060 A**, emergency **1,270 A (120%)**, day-2 loading **1,150 A (108%)**, and the SWEEP's 75 °C crossing at **1,060 A** with 1,150 A reading 89 °C once corrected. Then the drifted sensor makes the corridor look *cooler* than it was, which is the story the campaign tells; at present the SWEEP implies the corridor could carry more than believed, which inverts it.
BL-05	CLOSED	`interiors:` panel rows, six values	Thousands separators have been split by a space: `'6, 400 MW'`, `'4, 200 MW·s'`, `'1, 060 A'`, `'1, 240 A for 30 min'`, `'14, 200'`, `'6, 900 MW'`. These render on the room's instrument panel exactly as written, so the player reads "6, 400 MW" on the wall.	Remove the spaces: `6,400 MW` etc. Worth grepping `interiors:` in every book for `', '` inside a quoted numeric value.
BL-06	**WORTH**	METER interior panel vs the day-3 TRACE	The panel says the SCADA scan is **4 s** and the TRACE says **2.0 s**; the panel says last calibration was **41 days** ago while the stake and the TRACE both say the relay clock was last checked **in the spring** (months). The panel is the instrument the room reasons with, so both numbers are read against the stop.	Set the panel to `2 s` and `last checked: spring` (or a month count consistent with it).
BL-07	**WORTH**	Day 4 TRIGGER "What you do before a relay does it for you"	The stake gives the room "about 40 minutes" before the relay acts, and the TRIGGER's stream runs over **24 days** with `hoursLeft` from 576 to 0 and a six-hour redispatch lead time. Two incompatible clocks for one decision: minutes in the prose, weeks on the panel.	Decide which the day is about. If it is the 40-minute relay margin, the stream should be minutes and the lead time a redispatch that fits inside it. If it is the multi-day thermal creep the SWEEP later re-reads, the stake should say so instead of "40 minutes". The second reads better against day 11.
BL-08	**WORTH**	~12 stakes and scenes	Numeral-normalisation damage, same class as three other campaigns: "3 things arrive together… Reyes has 1 control room, 1 crew" (day 2); "the lower-impedance circuit carries	Editorial pass restoring the word where the digit

			twice the current of the older **1**" (day 7); "the newer **1** is 3 ohms"; "**2** methods that do not share the corridor sensor" (day 11); "**1** corridor above continuous rating" (interior); "11 feeders are still dead and the crews can only be in **1** place at a time" (day 5 briefing).	replaced one that is not a count.
BL-09	TASTE	Day 13 "Merit order" (SCIENCETANK)	The allocation rules are authored as four numeric constraints in prose (≥80 committed, largest ≥35, unsupported <15, supported ≥20) and the `recommended` block gives exactly A:55 B:25 — the unique allocation satisfying all four with two proposals. So the "portfolio judgement" the background describes has one arithmetic solution. Defensible as authored, but it is a puzzle rather than a judgement.	If the intent is judgement, widen the bands (largest ≥30, supported ≥15) so several allocations pass. If the intent is a puzzle, the background should stop describing it as a spread of belief.
BL-10	TASTE	~24 stops	The repeated format essays in `background`. Same as every campaign; noted again because this book's stop-specific backgrounds are the best in the repo (the current-transformer explanation, "what sets the temperature", "why the room wants the wrong thing" on black start) and they sit behind boilerplate the player has learned to skip.	Keep each essay on first use per format.

Day-by-day notes (short)

- **Day 1** — Round-trip efficiency, the transformer stop (BL-01) and Kirchhoff's node law. A good opening triple, one broken key.
- **Day 2** — DELEGATE with "watch it" explicitly refused is the format used properly. The SEQUENCE with authored `axis` / `ends` ("Improves your information" → "Cannot be taken back") is a model non-chronological rail.
- **Day 3** — The df/dt inference (300 MW from a slope and an inertia figure) is exactly the right hard question. The TRACE on three records of eight seconds is excellent and the relay-clock answer is derivable.
- **Day 4** — BL-07 sits here. The physics of the TRIGGER is right; its clock is not.
- **Days 5–7** — Distribution block. The dialysis-clinic CHOICE is decided on two grounds (consequence of waiting *and* job length) and says so, which is better than most triage questions.
- **Day 6** — BL-02. Otherwise the √3 stop and the synchronising SEQUENCE are both strong.
- **Day 8** — Fault current and the CONTROL on a current transformer (steady DC produces nothing) is a genuinely surprising, genuinely fair experiment.
- **Days 9–12** — The campaign's spine, and it holds. DEGENERACY on the island slope, CHAIN on black start, the sensor DIAGNOSIS, the SWEEP, and two CASEBOOKs that ask which conclusions survive a bad input. Nothing else in the catalogue does this.
- **Days 13–15** — SCIENCETANK (BL-09), the annealing CHOICE, and an ATTEST that spends two checks by consequence. The closing CHOICE names the practice the whole fortnight was about.

Closing blurb

Three paragraphs. The middle one — "What it cost… What is unfinished: the trip has no proven cause, the store has been measured once, and the practice that would have caught all of it is a paragraph in a report until somebody on nights makes it a habit" — is the best paragraph of prose in any of these games. The third addresses the player specifically. Keep all three.

Warm-ups

All seven authored and specific: six earths against a log that says five; the contractor who wants a verbal instruction in the yard; catching Whitlock before she drives off with the only unfiled relay settings. Names arrive with jobs. No findings.

What to keep

- The wrong-number spine (day 4 → day 9 → day 11 → day 15). It is the best long-form argument in the repo.
- Day 11's two CASEBOOKs. "One wrong number voids what leaned on it and nothing more" is a lesson no other campaign teaches.
- Day 10 as a deliberately quiet shift.
- The corridor as a recurring character — once BL-04 makes its numbers agree.